SUSTAINABLE INDUSTRY SPRINT

Type : Group (4 members)

Duration : 3.5 - 4 hours

"Design the Future. Solve for Sustainability."

EVENT OBJECTIVES

- Expose students to real industrial sustainability challenges.
- Develop structured problem-solving under time constraints.
- Improve professional communication and decision-making.
- Encourage interdisciplinary collaboration and strengthen presentation.

⏳ EVENT FLOW & TIMELINE

🟢 0:00 – 0:20 | Registration & Orientation

What Happens:

• Team verification

• Distribution of A4 problem analysis sheets

• Rules & evaluation criteria explained

• Judges introduced (Faculty / Alumni / Industry Guest).

🟢 0:20 – 0:30 | Problem Statement Reveal

• One problem sheet is revealed to all teams.

• All teams receive identical data sheets.

• No prior preparation allowed.

Teams must work within given:

- Budget constraints
- Timeline limits
- Technical restrictions

🟢 0:30 – 2:00 | Analysis Phase (90 Minutes)

Teams must:

✓ Clearly define the problem

✓ Identify root causes

✓ Brainstorm 2–3 solutions

✓ Select the best solution

✓ Estimate cost

✓ Analyze sustainability impact

✓ Assess risks and feasibility

✓ Develop implementation timeline

Participants record their structured thinking on provided A4 sheets.

🟢 2:00 – 3:00 | Presentation Preparation (60 Minutes)

Teams prepare a 5–7 slide PPT including:

- Problem Summary
- Key Insight
- Proposed Sustainable Solution
- Cost Breakdown
- Environmental Impact
- Implementation Plan
- Why This Works

🟢 3:00 – 3:50 | Boardroom Presentation Round

Each Team Gets:

• 7–8 minutes Presentation

• 5 minutes Q&A by Judges

Professional attire encouraged.

📜 RULES & REGULATIONS

Team Structure

• Exactly 4 members per team

• Interdisciplinary teams encouraged

Event Format Rules

🔹 Problem Statement

• Revealed only at event start

- No external prior preparation allowed
- Same dataset provided to all teams

🔹 Time Limits

- 90 min – Analysis
- 60 min – PPT Preparation
- 7–8 min – Final Presentation
- 5 min – Q&A

🔹 Permitted Resources

- Laptops allowed
- Internet usage: (Decide – Allowed / Restricted / Not Allowed)

🔹 Mandatory Solution Components

Each team must include:

- Clear Problem Definition
- Root Cause Analysis
- Sustainable Solution Proposal
- Cost Estimation
- Environmental Impact Assessment
- Implementation Timeline
- Risk & Feasibility Analysis

📄 SAMPLE PROBLEM STATEMENTS

PROBLEM 1

EV Battery Waste Optimization – Clean Mobility Startup.

🏢 **Company Background**
"VoltRide Mobility" is a mid-sized Indian EV startup manufacturing electric scooters. They produce 25,000 lithium-ion battery packs per year.

⚠️ **Current Issue**

- **10% of batteries fail quality tests.**
- **Disposal cost per rejected battery: ₹2,200**
- **Increasing regulatory pressure on battery waste.**
- **Customers demanding sustainability transparency.**

💰 **Constraints**

- **Budget limit: ₹75 lakhs**
- **Cannot change battery supplier immediately.**
- **Solution must be implementable within 12 months.**

🎯 **Target Outcome**

- **Reduce battery waste by at least 50%.**
- **Lower disposal cost.**
- **Improve sustainability reputation.**

PROBLEM 2

Sustainable Packaging Redesign – Food Manufacturing Company.

🏢 **Company Background**
"FreshBite Foods" produces packaged snacks sold nationwide.

Annual packaging usage:

- **1.5 million plastic packets**
- **Cost per packet: ₹4.20**

⚠️ **Current Issue**

- **Plastic waste criticism from consumers.**
- **Retailers asking for eco-friendly packaging.**
- **Packaging must remain moisture-resistant.**

💰 **Constraints**

- **New packaging cost must stay under ₹5 per unit.**
- **Shelf life must remain 6 months minimum.**
- **Machines cannot be completely replaced.**

🎯 **Target Outcome**

- **Reduce plastic usage by at least 60%.**
- **Maintain product quality.**
- **Keep overall cost increase under 15%.**

PROBLEM 3

Campus Energy Reduction Strategy – Private Engineering College.

🏢 **Background**
A 3,000-student engineering college spends:
- **₹1.2 crore annually on electricity.**
- **40% of usage comes from classrooms and labs.**

⚠️ **Current Issue**
- **Rising electricity tariff (increase expected 20% next year).**
- **ACs and lights often left on unnecessarily.**
- **Administration wants visible sustainability initiative**

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💰 **Constraints**
- **Budget: ₹50 lakhs.**
- **Solar installation possible but limited rooftop space.**
- **Cannot disrupt academic schedule.**

🎯 **Target Outcome**
- **Reduce electricity bill by 25% within 18 months.**
- **Improve campus sustainability ranking.**
- **Achieve measurable carbon reduction.**

REQUIREMENTS

📄 **Materials**

- Printed A4 Problem Sheets.
- Analysis Recording Sheets.
- A room with a projector- In order for the participants to explain/showcase their solution.

👩‍💻 **Participants**

- Must bring laptops.
- Formal or semi-formal attire recommended.

🎖️ **Rewards**

Prizes for the winning teams.

🏆 1st Prize – Trophy + Certificate.

🥈 2nd Prize – Certificate.

🌿 Sustainability-themed badges/stickers.